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Policy Recommendations for Higher Education Institutions to Begin Advancing from Digital Transformation to Bifurcation

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Abstract

Background Higher education institutions, particularly universities of technology, are ideally situated to advance critical inquiry and implementation of Responsible Research and Innovation (RRI) and digital transformation. This is a fundamental mission of higher education and universities of technology, to act as generators and incubators of technological innovation and educating and training the innovators of tomorrow in responsible and ethical conduct. In this brief communication, we will set out to outline policy positions for actors (at policy-making level as well as those in university management and governance) in higher education institutions to consider in order to support responsible and sustainable digital transformation. Moreover, here we outline our position that it is time for a paradigm shift in digital transformation that attempts to separate it from neoliberal economic models and mindsets, and looks to alternative epistemologies and political economies (and ontologies)

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for ecologically sustainable and ethical digital transformation. Here we propose a concept of bifurcation, or a radical break or fork, or series of forks, from the deleterious and unsustainable practices and systems of the past that have led to a global ecological emergency. We propose to use an interdisciplinary notion of bifurcation in order to underline the need to facilitate and foster a radical change of recent culture and late-modern societies – supported and enabled by the best digital technologies that we do have and fostering also the inception and implementation of innovative new technologies fit for the purpose of this bifurcation.

Keywords Bifurcation · Digital Transformation · Sustainability · Ethics · Epistemology · Responsible Innovation · Education

Introduction

The late 20th and early 21st centuries have witnessed the radical growth and proliferation of digital technologies and their applications across multifarious, even ambient[1] contexts—where processes within diverse fields from healthcare to education, industry, commerce, government and democratic participation are being augmented or entirely re-conceived or replaced through ground-breaking innovations. Indeed, novel digital technologies have the capacity to catalyse paradigm shifts and transform social, political, economic, educational and research contexts. Taking the example of social media alone, we can observe significant digitally mediated changes in how many of us spend our spare time, do business, and relate to each other due to the new affordances offered by these platforms (Facebook, X, etc.). Recent advances in the field of Artificial Intelligence including the field of large language models and (especially) generative AI, are swiftly changing the parameters of many aspects of our lives, both in work and recreation. Such advances are pharmacological in nature having both desirable and undesirable implications [1, 2]. They can be conceived as ambivalent or ambiguous at the core.

Digital technologies motivate the creation and alteration of new and existing social and societal practices. They generate transformative change and an epochal break that can concretely influence the course of development of individuals, institutions and society towards uncertain futures. The accelerated transformation is ongoing, ambiguous, and characterised by the introduction and evolution of digital technologies and their applications or uses from novel appropriations. The speed of technological change causes instability as it is far beyond society's capacity to integrate the technologies or technical objects into their cultural and social contexts.

Higher education institutions, particularly universities of technology, are ideally situated to advance critical inquiry into digital transformation and to facilitate and foster the implementation of a new wave of Responsible Research and Innovation (RRI) regarding digital technologies. It should be a fundamental mission of higher education institutions to act as generators and incubators of socio-technological innovation and cultural contextualization. This includes in particular educating and training the innovators of tomorrow in responsible and ethical conduct.

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[1] By "ambient" here we refer generally to technologies that operate in the background, such as in the case of the background human-technology relations identified by Don Ihde, where our experiences are said to be shaped by technologies that are not necessarily observed by us but operate within the background [20, 21]. Today ambient computing technologies provide a timely example of this, where "[…] ambient computing refers to technology that is immersed in your surroundings, ready to help without any prompting" [22].

Whilst our interdisciplinary working group recognises the role that universities have in contributing to the future of digital transformation, it equally acknowledges their responsibility for realising an inclusive and equitable (cultural and ethical) transformation that is ecologically sustainable. The working group endorses research and advocacy on responsible and sustainable digital transformation as well as questioning the assumptions and motivations underpinning digital transformation initiatives (for example the development and implementation of digital platforms, systems and tools including those driven by artificial intelligence).

Digital technologies have a growing impact on the arts, science and research, in particular on the methodological and epistemological dimensions of research practice. AI is inducing a paradigmatic technoscientific shift to the science and research system. To some extent the traditional theory, knowledge and explanation orientation of modern science is being replaced by a more predictive and technological orientation: We are experiencing a change from science to technoscience[2] in the sphere of research and education. Digital skills seem to replace mathematical skills. This transformation needs an analysis, an assessment and critical shaping – as early as possible.

Our interdisciplinary working group, assembled from members of the European University of Technology's (EUt+) European Culture and Technology Laboratory (ECT Lab+) and European Sustainability Science Laboratory (ESSLab+), was established precisely to collaborate on investigating and advancing models of ethically and ecologically sustainable frameworks for guiding digital transformation. This brief communication represents the initial output of this working group on digital transformation in proposing a new (normative) way of thinking about digital transformation, that is, not as a transformation but as a bifurcation, as well as policy recommendations that can tentatively support the shift to a new ethical and sustainable way of approaching the adoption and evolution of digital technologies in the university. There is an urgent need for advancing and institutionalizing a framework for responsible digital transformation in higher education.

This brief communication begins by introducing the EUt+, the ECT Lab+, and the ESSLab+, and proceeds to elaborate upon a conceptualisation of bifurcation, before proceeding to initial policy recommendations.

Institutionalising Responsibility for the Future: The European University of Technology

The vision and mission of the European University of Technology (EUt+) are underpinned by the pivotal role that technology plays in forging an inclusive and sustainable future: humanity today faces challenges of unprecedented breadth such as climate change, overused resources, growing inequality, threats to democracies, and the social consequences of the digital era.

Responding to these challenges necessarily involves technology, but such responses need to be multi-faceted. They must take into account the needs and aspirations of people and be attentive to our environment, they must be respectful of individual freedom and diversity. This can only be achieved by empowering citizens to act responsibly with respect to technology, and researchers who fully comprehend the potential of technology as well as the risks of neglecting its purpose. This requires a fundamentally new approach to technology and the training and education of people to foster it. It requires a new model of university.

The European Culture and Technology Laboratory (ECT Lab+) is a pan European inter- and transdisciplinary research institute within the European University of Technology. The ECT Lab+is carrying out research into the relation between technology and culture in all its forms. The ECT Lab+mobilizes the concept of technē to encapsulate the relation between all forms of techne as arts, craft and technics. The ECT Lab+promotes the concept of technology as the logos of technics to include all dimensions of discourse about technē, including mathematics, engineering, informatics—and also from cultural, social and political discourses about technics and techne. Technological innovation does not take place in a vacuum, it takes place in a locality with and for society, hence the concept of bifurcation is exploring the ethical, aesthetic and ecological questions of technology. Since its founding, the ECT Lab+has addressed numerous questions surrounding the relationship between society and technology in work which informs its understanding of what a bifurcation is or can be [3–11].

The European Sustainability Science Laboratory (ESSLab+) is a multinational, multi-campus,

[2] This term "[…} indicates the complex interactions between contemporary science and technology, that have become practically inseparable" [23]. See also [24].

trans- and inter-disciplinary research group. Its aim is to contribute to the research effort in sustainability sciences. It fosters innovation in sustainability through critically examining socio-technical transitions required to achieve sustainable lifestyles in feasible timeframes, with a focus on operating within planetary boundaries. In order to achieve this, the inter-disciplinary aspect of the lab is crucial; researchers with diverse backgrounds are necessary to remove ambiguity when assessing complex, multi-level problems, and methodological rigour can be improved through epistemological agility. The core missions of the lab are the production of knowledge on sustainability, the transfer of knowledge to impact society, and the discovery and exploration of ways to practice research in a more sustainable way. In essence, the ESSLab + promotes a paradigm shift towards a more holistic ecological view. As the importance of sustainable transformation through policy and scientific advancement increases across Europe, the ESSLab + is poised to significantly contribute to EUt + activities and the broader adoption of sustainability-informed policies and practices.

Thinking beyond Transformation—and the Need for Bifurcation

Here we will outline our position that it is *time for a paradigm shift in digital transformation that attempts to separate it from neoliberal economic models and mindsets, and looks to alternative political economies and epistemologies (and ontologies) for ecologically sustainable and ethical digital transformation*. Here we endorse a concept of *bifurcation*, or a radical break or fork, or series of forks, from the deleterious and unsustainable practices and systems of the past that have led to a global ecological emergency and to a variety of crises [12]. In other words, the societal grand challenges are to be recognized as challenges for the academy and the university system.

It is time to start thinking beyond "transformation" and towards "bifurcation". The need for more radical change has been seen through groups like Extinction Rebellion and also by the Scientists for Future, but has been seen more recently with the movement of students in the French engineering school AgroParisTech. These students demonstrated by pleading–during their graduation ceremony from the prestigious university–for a bifurcation from the university's curriculum and training for careers in socially and environmentally destructive agribusiness industries [13]. The students chose to desert careers that they decided were unethical by their very nature, and their speeches were necessarily an indictment against a university and education system that prepares students to enter such careers, leveraging ethically non-neutral knowledge and technologies and advancing technological innovation uncritically in the service of industry, and thereby maintaining systems of exploitive and destructive practices and power relations. Increasingly, what students are demanding is a radical change to the way they are taught technological and engineering skills, asking that the curriculum focus on questions of the ethical and ecological impacts of technology, and for alternative models of living and working that do not contribute to a declining social, political and environmental ecology.

Bifurcation can mean challenging the present-day late-modern mindset, in particular the way of conceptualizing, thinking and framing humans, societal human-nature relations as well as societal human-technology relations. The established human-centric perspectives – interlaced with the neoliberal profit-maximizing *homo economicus* and the instrumentalist view of technology – that induced the problematic Anthropocene and its grand challenges, need to be revealed, scrutinized, and revised. Novel processual and inherently critical concepts should be developed in order to relate humans, nature, technology, society and economics in a different, namely in a relational way, beyond various reductionist takes on them. And, in a broader sense, to bifurcate can also mean to take different (onto-) epistemological paths based on transdisciplinary crossovers between embodiment, sciences, and the arts. This approach draws from various sources, including Lynn Margulis' work on symbiogenesis, new materialist feminist theories [14–17], and earth-bound cosmologies (Sumak

Kawsay,[3] Whakapapa,[4] Celtic cosmology[5] etc.).The goal is to challenge the dominant Western-centred narratives and mindsets of the Anthropocene, which are built on artificial binaries and a stable notion of climate change. From this perspective, the dominant human-nature- and the societal nature-relations must be questioned and scrutinized. To achieve this bifurcation, the suggestion is to incorporate knowledge from diverse worldviews and disciplines.

We propose to use an interdisciplinary notion of bifurcation in order to underline the need to facilitate and foster a radical change of recent culture and late-modern societies – supported and enabled by the best digital technologies that we do have and fostering also the inception and implementation of innovative new technologies fit for the purpose of this bifurcation. We therefore propose thinking and acting around two kinds of bifurcation, a cultural/societal one, and a digital one that emerges from and can support or sustain the endeavour of the larger cultural shift. The conceptual approach of the working group assumes that the new developments, correlated with AI, will not proceed according to so-called technological determinism. According to this paradigm of determinism of technological development, the digital transformation would not be sufficiently shapeable, but only adaptation and adjustment could be developed in response to it. The approach of the working group disputes this view, which can be described as "technology-driven". Rather, it is assumed that technological development can be fundamentally shaped by society, culture and politics as well as by development and usage practices.

Responsible digital bifurcation (which could be understood as a futuring [18] and more symbiotic technological development) can only emerge from and in association with a larger societal bifurcation. Such a bifurcation can fruitfully begin within universities as sites of inter- and transdisciplinary critical inquiry and participatory experimentation, and where programmes of "reworlding" [18] can be undertaken. By reworlding we mean work on critiquing and redesigning the institutions (broadly construed) and technologies that entrench and motivate unsustainable, unethical and exploitive practice—where work on envisioning alternative futures with reformed or reforged systems of incentives can be undertaken. The total assemblages in which technological transformation takes place require redesign, and such redesign can be supported with technological tools—society and technology have a circular or interactional influence and one which researchers and educators can actively participate in, in dialogue with wider society across diverse localities.

In our working group, it is a matter of course that we practice an integrative and inclusive scientific culture that is characterized by a two-culture or a big interdisciplinary approach [19]. This places the cultural bifurcation of society as a whole in an ethical, societal, social and economic context.

Policy Recommendations

Here we propose general policy actions to be explored and implemented within the context of higher education institutions by local, national, and international stakeholders with responsibility for shaping education and research strategies in higher education, in order to

[3] Sumak Kawsay is the Indigenous cosmology from people's of South America, particularly Ecuador and Bolivia. Sumak Kawsay is an ecocentric cosmology (as opposed to an anthropocentric one) raising a holistic vision of development in coexistence with ecosystems as the central axis of harmony with life and the activities of human populations in different territories. Quechua philosopher Javier Lajo makes a distinction between Sumak Kawsay (Buen Vivir) and the Western conception of Living Well (Aristotle) that privileges "thinking" above "feeling", conditioning science and technology to the principle of reason or the "logos", resulting in the separation and domination of the subject over the object and of man over nature. Sumak Kawsay is about responsibility towards the planet, without hierarchies of domination, encompassing epistemic diversity (thinking-well) and care (doing-well) [25, 26].

[4] Whakapapa is a foundational concept in Māori culture. Although it has been translated to English as 'genealogy' or 'lineage' its meaning is deeper and broader than the English translation suggests. Whakapapa is a framework that connects all living and non-living things, known and unknown in both physical and spiritual worlds. Whakapapa organises relationships, histories, philosophies and spiritual beliefs ensuring that knowledge is preserved and transmitted across generations [27].

[5] Celtic cosmology has been reconstructed by Celtic Studies scholars through the analysis of early medieval Irish and British texts, and archaeological and comparative evidence. Celtic cosmology is consistent with animistic traditions attributing sacred qualities to nature and landscape. Deities and supernatural beings were associated to natural phenomena and the cosmic order. Their stories reflecting concern about astronomical events, sovereignty, kingship, and the maintenance of the cosmic balance [28].

facilitate progress on responsible digital and societal bifurcation. These steps are tentative ones towards a necessary bifurcation, for now laying the foundation for a reorientation and re-evaluation of values and their implementation in university and society that can support a more radical shift in priorities going forward. Such recommendations can position universities as leaders and exemplars in promoting positive change in how they operate, and as thought leaders shaping the minds of tomorrow's innovators, digital citizens and technology end users. Universities should:

Hardware and Software Support in the University

- Lead by example by promoting and implementing sustainable practices in the procurement, usage and disposal of electronic devices, with a focus on the management of e-waste and the promotion of circular economy and right-to-repair principles.
- Adopt procurement policies prioritising repairable, upgradeable, and long-lifespan hardware, reducing electronic waste and financial overheads caused by frequent replacements.
- Promote the integration of green algorithms policies for sustainable digital technologies in both academic and research settings.
- Prioritise twin transition integration through the adoption of digital infrastructure that is advanced from a functional standpoint while being energy-efficient and informed by circular economy principles, including ease of repairability and recyclability.
- Integrate green algorithms policies into academic and research practices involving digital technologies, emphasising an "optimise first, upgrade later" approach for both software development practices and data processing.
- Prioritise open-source software and digital tools as the default choice for their digital infrastructure. Public funding should explicitly require developing, using, and maintaining open-source alternatives over proprietary systems to ensure digital sovereignty and interoperability.
- Promote Low-Tech computing principles in university IT policies, including lightweight software, modularity, and long-term platform stability, to reduce energy and resource consumption. Institutions should introduce Minimal Computing Guidelines to ensure that software and digital services prioritise efficiency, adaptability, and longevity over continuous hardware upgrades.
- Guarantee access to new technologies (assessed to be socially responsible) such as social networks and AI for staff and students. This includes technical availability as well as the rules for use (e.g. rules for AI tools based on a common understanding).

Education and Skills

- Increase investment in digital skills (including broadly in digital, AI and media literacy), retraining and upskilling for evolving and multifarious digital professions and practices—inclusively and non-discriminatively for learners of diverse backgrounds and identities. Simultaneously, this should not be done at the expense of other essential skills necessary for a meaningful and productive life. Education should focus on skills for ethical and responsible professions, and not targeting market needs at the expense of social responsibility and sustainability.
- Support the development of skills to analyse and assess the inherent ambiguity of any kind of digital technology in the individual lifeworld and build capacities to enable an adequate, critical and reflexive usage and shaping of digital technologies.
- Invest in education, training and resources to support students and researchers in the development of sustainable digital technologies utilising ethically sourced material components and labour.
- Improve access to digital, technological, and ecological ethics education to all disciplines in third level education, understanding that technological tools and practices are not value neutral and touch all industries and professions. Make such education programmes, especially those with a focus on responsible research and innovation, mandatory in STEM disciplines, recognising the role of STEM graduates in driving digital transformation forward in developing innovate new technologies and applications.

- In particular, note the great opportunities and risks posed by artificial intelligence systems and applications, and the corresponding need to facilitate the cultivation of critical thinking and ethical deliberation in students and researchers who will be responsible for future breakthroughs in AI, which could have profound societal impacts.
- Note that such digital, technological, and ecological ethics programmes should emphasise that the digital is not post-material, and should instruct learners on the material aspects of socio-technical systems and (often exploitive and extractive) supply chains (including related underlying power relations and interest groups).
- Champion equality, diversity and inclusion in the design, adoption, implementation and deployment of digital technologies and incorporate such approaches into RRI education for staff and students.
- Adhere to the Sustainable Development Goals (SDGs) in the adoption and design of digital technologies but also facilitate inquiry and experimentation into approaches beyond sustainable development stemming from dominant ontologies, seeking alternative pathways to sustainable futures with technology, rooted in different cultural movements.
- Foster and institutionalize opportunities of critique and critical engagement of and with digital technology—and implement critique and critical thinking about digital technologies in all curricula. This includes classes on digital ethics, technology assessment, ethical impact assessments, risk assessments and foresight methodologies.

Digital Governance

- Establish open-source governance models to oversee digital tools and infrastructure's ethical, transparent, and sustainable deployment.
- Enable a technological development and shaping of digital technology according to fundamental democratic values and participatory decision making.
- Mainstream various kinds of technology assessment and foresight methodologies into the lifecycles of digital transformation projects and initiatives, including ethical impact assessments, risk assessments, prospective technology assessments, and fundamentally support the development and implementation of such methodologies that can respect plural global and local value systems in RRI.
- Apply technology and impact assessments to their own digital transformation initiatives, evaluating whether proposed new digital and technological tools and processes add value to the university or displace meaningful human skills and competencies, or have other undesirable impacts (including adverse environmental impacts).
- Develop clear policies to ensure that institutional data is not exploited by third-party corporations, particularly in training AI and LLM technologies.
- Foster digital well-being and mental health awareness. Incorporate guidelines that address the psychological impacts of digital technologies, promoting balanced digital engagement and mental health support for students, staff, and researchers.

International Collaboration

- Continue to develop and implement sustainable digital technologies to enhance borderless virtual education and promote cross-border research and innovation, thereby also facilitating less dependence on carbon intensive international travel.
- Support (and fund) interdisciplinary and inclusive research collaboration (including contributory research) between the Global North and South on the topic of sustainable digital bifurcation. Interdisciplinary and cross-border research bringing together STEM and AHSS should focus on the study of just digital transitions or a just bifurcation, and the plurality of visions of just futures and how to arrive at them

Concluding Remarks

Higher Education and in particular universities of technology have a central role to play in the development of technologically responsible citizens. Perhaps the development of new technologists who see the role of ethical and ecological responsibility as their key concern will lead to the development of new technologies of the future which reflect values including respect for human dignity, freedom, democracy,

equality, the rule of law, and respect for human rights. The next steps of this interdisciplinary working group will be the development of self-reflexive tools for technological impact assessment.

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Declarations

Competing interests Authors have no conflicts of interest or competing interests to declare.

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